

Theorem (Jensen's Inequality). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be convex; i.e. for every $x, y \in \mathbb{R}$ and every $t \in [0, 1]$,

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y). \quad (C)$$

Let $x_1, \dots, x_n \in \mathbb{R}$ and let weights $\lambda_1, \dots, \lambda_n$ satisfy $\lambda_i \geq 0$ and $\sum_{i=1}^n \lambda_i = 1$. Then

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i). \quad (J)$$

Proof. We argue by induction on n .

Base case $n = 2$. Take $t = \lambda_1$ (hence $1 - t = \lambda_2$). Applying the convexity property (C) to the pair (x_1, x_2) gives

$$f(\lambda_1 x_1 + \lambda_2 x_2) \leq \lambda_1 f(x_1) + \lambda_2 f(x_2),$$

which is exactly (J) for $n = 2$.

Inductive hypothesis. Assume that for some integer $k \geq 2$ the inequality holds for every collection of k points with non-negative weights summing to one; i.e.

$$f\left(\sum_{i=1}^k \lambda_i x_i\right) \leq \sum_{i=1}^k \lambda_i f(x_i) \quad (\lambda_i \geq 0, \sum_{i=1}^k \lambda_i = 1). \quad (IH)$$

Inductive step (from k to $k+1$). Let $\lambda_1, \dots, \lambda_{k+1} \geq 0$ with $\sum_{i=1}^{k+1} \lambda_i = 1$. Define

$$\mu := \sum_{i=1}^k \lambda_i \in [0, 1], \quad \alpha_i := \frac{\lambda_i}{\mu} \quad (i = 1, \dots, k)$$

when $\mu > 0$; when $\mu = 0$ the statement is trivial because then $\lambda_{k+1} = 1$ and (J) reduces to $f(x_{k+1}) \leq f(x_{k+1})$.

For $\mu > 0$ each $\alpha_i \geq 0$ and $\sum_{i=1}^k \alpha_i = 1$. Set

$$y := \sum_{i=1}^k \alpha_i x_i.$$

Application of the inductive hypothesis. Since the α_i form a convex weight vector for the first k points, (IH) yields

$$f(y) = f\left(\sum_{i=1}^k \alpha_i x_i\right) \leq \sum_{i=1}^k \alpha_i f(x_i). \quad (1)$$

Rewrite the whole combination as a two-point convex combination. Observe that

$$\sum_{i=1}^{k+1} \lambda_i x_i = \mu \left(\sum_{i=1}^k \alpha_i x_i \right) + (1 - \mu) x_{k+1} \quad (1)$$

$$= \mu y + (1 - \mu) x_{k+1}. \quad (2)$$

Because $\mu \in [0, 1]$, (2) is a convex combination of the points y and x_{k+1} .

Apply convexity to (2). Using (C) with $t = \mu$, $x = y$ and $y = x_{k+1}$,

$$f\left(\sum_{i=1}^{k+1} \lambda_i x_i\right) = f(\mu y + (1 - \mu)x_{k+1}) \quad (2)$$

$$\leq \mu f(y) + (1 - \mu)f(x_{k+1}). \quad (3)$$

Insert the bound for $f(y)$ from (1). Multiplying (1) by μ gives

$$\mu f(y) \leq \mu \sum_{i=1}^k \alpha_i f(x_i) = \sum_{i=1}^k \lambda_i f(x_i). \quad (4)$$

Also $1 - \mu = \lambda_{k+1}$, so $(1 - \mu)f(x_{k+1}) = \lambda_{k+1}f(x_{k+1})$. Insert (4) into (3):

$$f\left(\sum_{i=1}^{k+1} \lambda_i x_i\right) \leq \sum_{i=1}^k \lambda_i f(x_i) + \lambda_{k+1}f(x_{k+1}) \quad (3)$$

$$= \sum_{i=1}^{k+1} \lambda_i f(x_i), \quad (4)$$

which is precisely (J) for $n = k + 1$.

Completion of the induction. The base case $n = 2$ holds, and the inductive step shows that truth for k implies truth for $k + 1$. By the principle of mathematical induction, (J) holds for all integers $n \geq 2$. The case $n = 1$ is immediate because both sides equal $f(x_1)$.

Hence

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i).$$

□